

60/40 in 2022: When the Stock-Bond Correlation Flipped

During the 2022 inflation shock, the balanced portfolio failed because its diversifier stopped diversifying.

For two decades, the 60/40 portfolio rested on one quiet assumption: **when stocks fall, bonds rally**. In 2022 that assumption broke. Stocks fell ~18%, long Treasuries fell ~31% — the worst year for a US balanced portfolio since the Global Financial Crisis, and by some measures since the 1930s. This case study reconstructs what happened, why the correlation flipped, and what it means for every portfolio built on the stock-bond hedge.

PROJECT CARD

CATEGORY

Portfolio Optimization

LIBRARIES

Pandas · statsmodels · seaborn

ASSETS

SPY + AGG (+ TLT for the long-duration pain)

TIMEFRAME

Jan 2000 – Dec 2023, focus 2022

USE

Stress-testing the diversification assumption every balanced portfolio silently relies on

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
plt.rcParams["figure.dpi"] = 110
sns.set_style("whitegrid")
```

1. The setting: two decades of a free hedge

From roughly 2000 to 2021, US stock and Treasury returns were **negatively correlated**. Every equity selloff — 2008, 2011, 2018, March 2020 — saw bonds rally as investors fled to safety and the Fed cut rates. A 60/40 investor got equity upside with a built-in shock absorber, and the strategy compounded through every crisis.

The subtlety everyone forgot: that negative correlation is a **regime**, not a law. It held because inflation was low and stable, so growth shocks dominated — bad news for stocks was good news for bonds (rate cuts coming). In the 1970s–90s, when *inflation* shocks dominated, the correlation had been **positive**.

2. The event: 2022 month by month

Inflation, dismissed as “transitory” in 2021, printed 7%+ into 2022. The Fed delivered the fastest hiking cycle in four decades — from 0.25% to 4.50% in ten months. Rising yields crushed bond prices (see our duration tutorial: TLT’s

~17.5-year duration × ~2% yield rise ≈ −35%), while the same rate shock compressed equity multiples.

Both engines of 60/40 stalled at once. There was nowhere to hide inside the classic mix.

3. The mechanism: why inflation flips the sign

- **Growth-shock regime** (2000–2021): bad economy → stocks fall, Fed cuts → bonds rise. Correlation **negative**. 60/40 self-hedges.
- **Inflation-shock regime** (1970s, 2022): inflation up → Fed hikes → *discount rates rise for everything* → stocks

and bonds fall together. Correlation **positive**. The hedge becomes a second exposure to the same risk.

One variable — which type of shock dominates — determines whether bonds protect you or double your bet.

4. The evidence

4.1 The data

SPY (S&P 500), AGG (aggregate bonds) and TLT (long Treasuries) from AGG’s inception through 2023.

```
TICKERS = ["SPY", "AGG", "TLT"]
START, END = "2003-10-01", "2023-12-31"

def load_prices(tickers, start, end):
    """Adjusted-close prices: yfinance first, Stooq as fallback."""
    try:
        import yfinance as yf
        df = yf.download(tickers, start=start, end=end, auto_adjust=True, progress=False)["Close"]
        if not df.empty:
            return df[tickers].dropna()
    except Exception as exc:
        print(f"yfinance failed ({exc}); trying Stooq...")
    cols = {}
    for t in tickers:
        url = f"https://stooq.com/q/d/l/?s={t.lower()}.us&i=d"
        cols[t] = pd.read_csv(url, parse_dates=["Date"], index_col="Date")["Close"].rename(t)
    return pd.concat(cols, axis=1).loc[start:end].dropna()

px = load_prices(TICKERS, START, END)
rets = px.pct_change().dropna()
print(f"{len(px)} trading days, {px.index[0].date()} → {px.index[-1].date()}")
```

4.2 Annual returns: 2022 in context

The one chart that tells the story — find another year where *both* bars are deeply negative.

```
annual = (1 + rets).resample("YE").prod() - 1
annual.index = annual.index.year

ax = annual[["SPY", "AGG"]].plot(kind="bar", figsize=(11, 4.5),
                                color=["steelblue", "darkorange"], width=0.8)
ax.axhline(0, color="black", lw=0.8)
ax.set_ylabel("Total return"); ax.set_title("Annual returns: stocks (SPY) vs bonds (AGG)")
ax.legend(["SPY (stocks)", "AGG (bonds)"])
for yr, row in annual.iterrows():
    if yr == 2022:
        ax.axvspan(list(annual.index).index(yr) - 0.5, list(annual.index).index(yr) + 0.5,
                    color="crimson", alpha=0.10)
plt.tight_layout(); plt.show()

print(annual.loc[[2008, 2020, 2022]].round(3))
print("\n2008 & 2020: bonds cushioned the crash. 2022: they amplified it.")
```

4.3 The correlation flip

Rolling 1-year correlation of daily stock and bond returns. Two decades below zero — then 2022.

```
roll_corr = rets["SPY"].rolling(252).corr(rets["AGG"])

fig, ax = plt.subplots(figsize=(11, 4.5))
ax.plot(roll_corr, color="steelblue", lw=1.2)
ax.axhline(0, color="black", lw=0.8)
ax.axvspan(pd.Timestamp("2022-01-01"), pd.Timestamp("2023-01-01"), color="crimson", alpha=0.12)
ax.set_title("Rolling 1-year stock-bond correlation (SPY vs AGG)")
ax.set_ylabel("Correlation")
plt.tight_layout(); plt.show()

print(f"Average correlation 2004–2021: {roll_corr.loc[:'2021'].mean():+.2f}")
print(f"Peak correlation in 2022–23: {roll_corr.loc['2022':].max():+.2f}")
```

4.4 The damage: 60/40 drawdown

A monthly-rebalanced 60/40 (SPY/AGG). Compare the 2022 drawdown with the GFC — and note how much *faster* 2022 hurt, because nothing offset anything.

```
w = {"SPY": 0.60, "AGG": 0.40}
port_rets = (rets[list(w)] * pd.Series(w)).sum(axis=1) # daily, approx monthly rebalance
port_val = (1 + port_rets).cumprod()
drawdown = port_val / port_val.cummax() - 1

fig, ax = plt.subplots(figsize=(11, 4.5))
ax.fill_between(drawdown.index, drawdown * 100, 0, color="steelblue", alpha=0.6)
ax.axvspan(pd.Timestamp("2022-01-01"), pd.Timestamp("2023-01-01"), color="crimson", alpha=0.12)
ax.set_title("60/40 (SPY/AGG) drawdown")
ax.set_ylabel("Drawdown (%)")
plt.tight_layout(); plt.show()

dd_2022 = drawdown.loc["2022":"2023"].min()
dd_gfc = drawdown.loc["2007":"2010"].min()
ret_2022 = (1 + port_rets.loc["2022"]).prod() - 1
print(f"60/40 total return 2022: {ret_2022:+.1%}")
```

```
print(f"Max drawdown 2022-23: {dd_2022:+.1%}")
print(f"Max drawdown GFC 2008-09: {dd_gfc:+.1%}")
```

And the counterfactual that stings: a 60/40 built with **TLT** instead of AGG — more duration, more “hedge” — did *worse* in 2022, because the hedge asset itself was the epicenter.

```
for bond in ["AGG", "TLT"]:
    pr = 0.60 * rets["SPY"] + 0.40 * rets[bond]
    r22 = (1 + pr.loc["2022"]).prod() - 1
    print(f"60/40 with {bond}: 2022 return {r22:+.1%}")
```

5. Post-mortem

What failed

- **The model, not the math** — MVO and risk parity both treated the stock-bond correlation as a stable input. It's regime-dependent, and the regime variable is *inflation*.
- **Duration as a hedge** — in an inflation shock, duration is the *risk*, not the hedge (see the bond tutorial's 2022 ETF table).
- **Recency** — twenty years of negative correlation felt like a law of nature. The 1970s said otherwise all along.

What survived

- **Commodities and trend-following** had a banner 2022 — the diversifiers nobody wanted during the long bull market.
- **Short-duration bonds** (SHY) lost little — the failure was duration, not bonds per se.
- **The principle of diversification** — but across *risk regimes* (inflation vs growth), not just asset classes.

The lessons

1. Correlation inputs deserve the same stress-testing as returns — *conditional* on inflation regime, not unconditional averages.
2. A hedge that depends on a regime is a position on that regime persisting.
3. The fix isn't abandoning 60/40 — it's knowing which environments it insures against, and which it doesn't.

Related content

- **Bond Pricing, Duration & Convexity** — why TLT lost 31% (the mechanics behind this case)
- **Risk Parity from Scratch** — the strategy this correlation flip hurt most
- **Copulas & Tail Dependence** — modelling co-movement beyond a single correlation number